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Track 01 · LongevityLLM Benchmarking

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This track sits at the intersection of AI and aging biology. You will be working with a fine-tuned model built specifically for longevity science, and your job is to push its limits, find its gaps, and do something useful with it. If you care about how AI performs in biomedical research and want your work to end up in a journal, this is your track.

Criterion	Weight
Utility: Does the benchmark assess an LLM capability that genuinely matters for aging biology? Does the gap analysis focus on failure modes that would affect whether a real user could trust the model?	5 pts
Diversity: Are samples, phrasings, and answer formats varied enough to prevent shortcut learning? A "compound → senescence effect" benchmark dominated by rapamycin experiments fails on the data axis. An MCQ where the correct option is always "A" fails on the semantic axis. A single rigid format that cannot be turned into binary, ternary, regression, or set-generation variants fails on the format axis — a model that has never seen negative DEG results will ace "Is this gene overexpressed in condition X?" but will break down once "Not changed" becomes a possible answer.	5 pts
Retrieval resistance: How likely is a SotA model to have already seen the answer in its training corpus? Tasks based on PubMed abstracts score poorly. Tasks based on processed raw datasets or obscure repositories score well.	5 pts
Statistical rigor: Are class imbalances handled, is the chosen metric appropriate (F1, balanced accuracy, MAE, Jaccard...), and is a baseline reported so that the headline number is interpretable?	5 pts

Prize: \$1000 per team and a mention in a peer-reviewed publication by the Insilico Medicine research team.

The Task

Longevity Bench (LB) is a collection of datasets intended for measuring LLMs' ability to deduce phenotypes from low-level data. Most tasks in it focus on age and mortality prediction, life span experiments, and associations between genes and species lifespan.

You may access some tasks used to evaluate LLMs [here](#).

Currently explored domains:

- Epigenetics: GEO Datasets on DNAm arrays;
- Clinical biodata: NHANES;
- Transcriptomics: GTEx;
- Genetics: OpenGenes, SynergyAge, CellAge;
- Proteomics: public Olink datasets.

Task ideas:

- **Multispecies aging clocks**: LB is mostly human-centric. A task asking an LLM to predict if a dog, or other animal could be used to check if a model has generalized the concept of aging across different species;
- **Anti-aging target potential**: The repositories documenting genes involved in longevity. Which targets that can prolong human lifespan are strongly biased toward well-studied pathways? Is there a way to quantify the potential of a gene to slow down aging based only on omic data?
- **Mouse strain longevity**: MGI provides detailed annotations of murine strains, including their genetic background, phenotype and allele profiles. Can a model identify the effect of a mutation on murine lifespan?
- **Factual accuracy**: L-LLM can explain its decisions in terms of aging biology and by applying different aging clocks. But are these arguments true or just seem plausible? Are the CpG sites and genes really used by the clocks? Are they truly involved in the pathways L-LLM says they are?

Task specifications:

- All tasks should be split into train-test cohorts, based on covariates that prevent data leakage. For GTEx data — by patient ID, for GEO data — by GSE accessions, for proteins — for proteomic data — *in vivo* experiments — by species, for *in vitro* — by cell line. Even if the task is small, there should be multiple ways to split it.
- All tasks need to have verifiable ground truth.
- Preferred task formats: binary classification, multiple choice question (MCQ), ternary choice (Yes/No/None/Neither), pairwise comparison (choose sample A or B), regression, set generation.
- Freetext output format that requires expert/agent review is discouraged;
- Each task needs to have a formally defined statistic for measuring performance. F1, accuracy, MAE, Jaccard similarity etc;
- No prompt shall exceed 30K tokens after tokenizing with cl100k_base;

- File format for final submission: JSONL files with ChatML system-user-assistant interaction
- Tasks involving sequence modalities should be avoided. E.g. predicting the effect of a mutation on lifespan requires an LLM to operate with a DNA sequence.
- Minimal N prompts in a task is 50.

Extra credit: a training signal for reasoning

Once your benchmark produces a concrete performance metric, you have a substrate for a new question: does the model arrive at its answer through trustworthy reasoning, or does it land on the right answer for the wrong reasons?

Citing a gene that does not exist, attributing it to the wrong chromosome or pathway, or contradictions between its thinking trace and its final token are all failure modes that survive any final-answer check. We want a fast, programmatic signal that scores the trace, not just the final answer — checked against millions of times when used as a training signal for reasoning, and resistant to surface-level tricks. Over capitalized tokens to detect genes, for instance, fails the moment a model stops capitalizing. The moment we evaluate on murine and *C. elegans* genes whose symbols are not capitalized.

We will share access to a HuggingFace endpoint of Longevity-LLM (L-LLM), a fine-tuned GPT-4o checkpoint that outperforms much larger models on the LB tasks. Run L-LLM on the benchmark, collect its reasoning traces, and prototype a scoring function over them. Promising directions include: checking that mentioned genes, CpG sites, or strains exist and have the properties claimed; checking consistency between the thinking trace and the final answer; grounding claimed pathway mentions in known biology. The strongest submissions will describe a scoring function that is automatically and demonstrably correlated with biological correctness on a held-out set of traces from your benchmark.

- ▶ **Guide: How to connect to the hosted LongevityLLM endpoint and run the LongeBench evaluation task against it.**
- ▶ **Endpoint and access token**